

Title

Circadian Patterns of Intraocular Pressure and Tolerability Outcomes in Patients Treated With Sepetaprost or Latanoprost

Abstract

Intraocular pressure exhibits diurnal and nocturnal variation, and agents that effectively suppress pressure throughout the 24-hour cycle may confer clinical advantages. We conducted a secondary analysis of the efficacy and safety data collected at multiple time points across day and night in our treatment cohort. Participants treated with sepetaprost showed greater suppression of nocturnal intraocular pressure compared to latanoprost in a subset of measurements, with a nominal statistical difference noted at the 0.10 significance level. Diurnal pressure reduction was comparable between arms. Posttreatment assessment revealed that intraocular pressure remained below baseline even 36-48 hours after discontinuation in the sepetaprost group. Systemic and ocular tolerability was similar, with ocular surface discomfort and conjunctival hyperemia reported at similar frequencies. These exploratory circadian analyses suggest sepetaprost may offer benefits in overnight pressure control, though larger studies are needed to confirm this finding.